

Football ML Project

From Baseline to Breakthrough

What We Improved · What We Achieved · What's Holding Us Back

What This Report Covers

01

Where We Started

Original notebook – ad hoc preprocessing, manual encoding, a single dataset, and rounded metrics.

02

Problems Identified in Original Code

Leakage risks, no reusable pipelines, biased evaluation, limited feature depth.

03

What We Improved (Data + Modelling)

Canonical pipeline, cross-season data, sklearn pipelines, proper continuous regression evaluation.

04

Results vs. Baseline

Classification: up to +15 pp accuracy. Regression: lower MSE, higher R^2 . Full CV breakdown included.

05

What's Holding Us Back

Noisy labels (~30%), overlapping tactical roles, missing formation context, feature gaps across seasons.

Where We Started

Original Notebook — ML_project_code.ipynb

Single Dataset Only

Only top5-players.csv used. No 2024/25 season data integrated into training.

Ad Hoc Preprocessing

Inline string splits, manual dropna(), no reusable pipeline. Different scaler per model cell.

Manual Label Encoding

Separate LabelEncoder per column. Fit before train/test split — potential data leakage.

Rounded Regression

Predictions cast to int before computing MSE — artificially inflates error figures.

5 Models, No CV Discipline

RF, Decision Tree, SVM, LogReg, Neural Net tested without consistent cross-validation.

Accuracy Only

No Macro-F1 or per-class recall. Class imbalance invisible in reporting.

Repeated Boilerplate

StandardScaler re-instantiated in every model cell. No shared pipeline structure.

Baseline Numbers

RF Holdout Acc: 0.7567 | RF CV Mean: 0.7394 | LR MSE: 1.3892 | R^2 : 0.8625

What We Improved

Data & Preprocessing Pipeline

1

Season-Aware Dataset Fusion

Appended top5-players.csv + players_data-2024_2025.csv into combined_base. Only aligned shared columns kept — no leaky merges.

2

Canonical Field Normalisation

Nation, Comp, and primary Pos standardised through one canonicalize_common_fields() function called on every dataset before joining.

3

Derived Per-90 Features

Per-90-minute versions of key attacking and progression stats added. Season and source_file columns appended for full traceability.

4

Leakage-Free Detailed Experiment

detailed_2024_2025 built by dropping 76 duplicated metadata columns (Pos_stats_*, Nation_stats_*, Comp_stats_*) that echo the target label.

5

Proper sklearn Pipeline Architecture

Median imputation → OneHotEncoding → model-specific scaling all inside a single Pipeline. No state leaks between train/test.

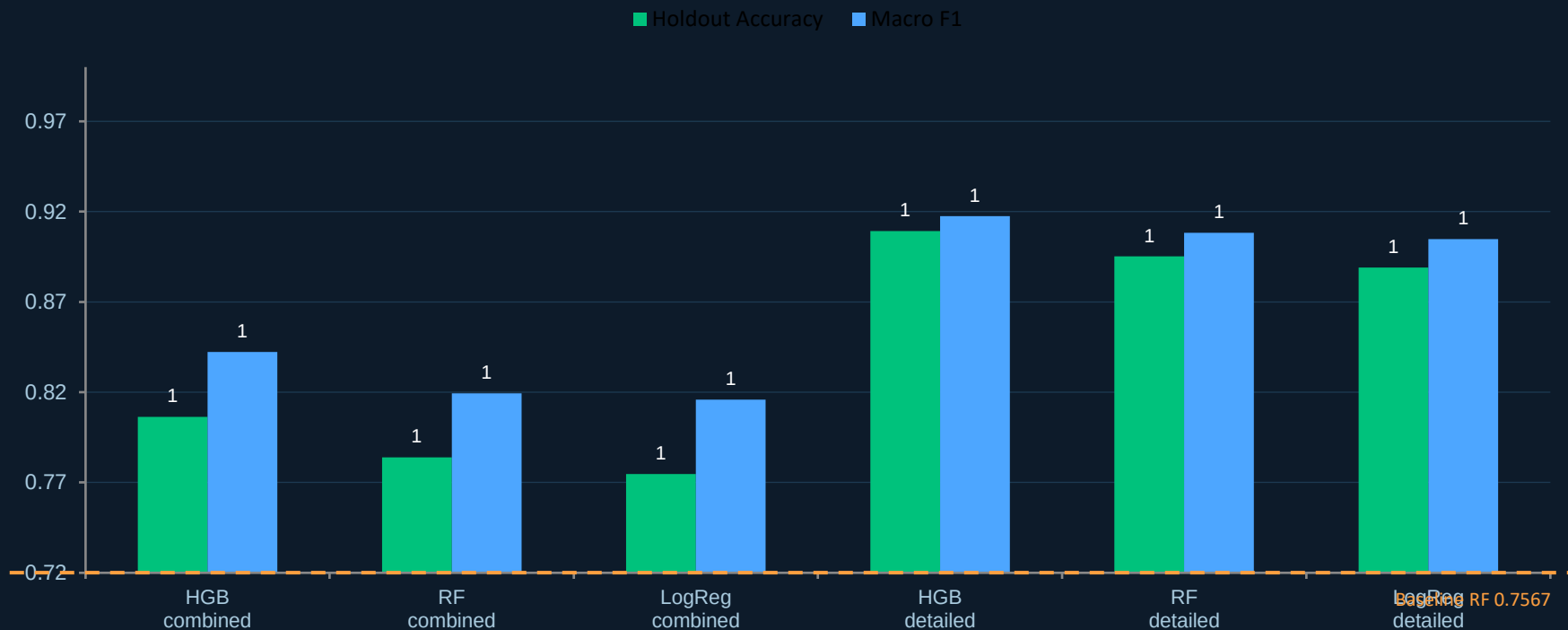
What We Improved

Modelling & Evaluation Strategy

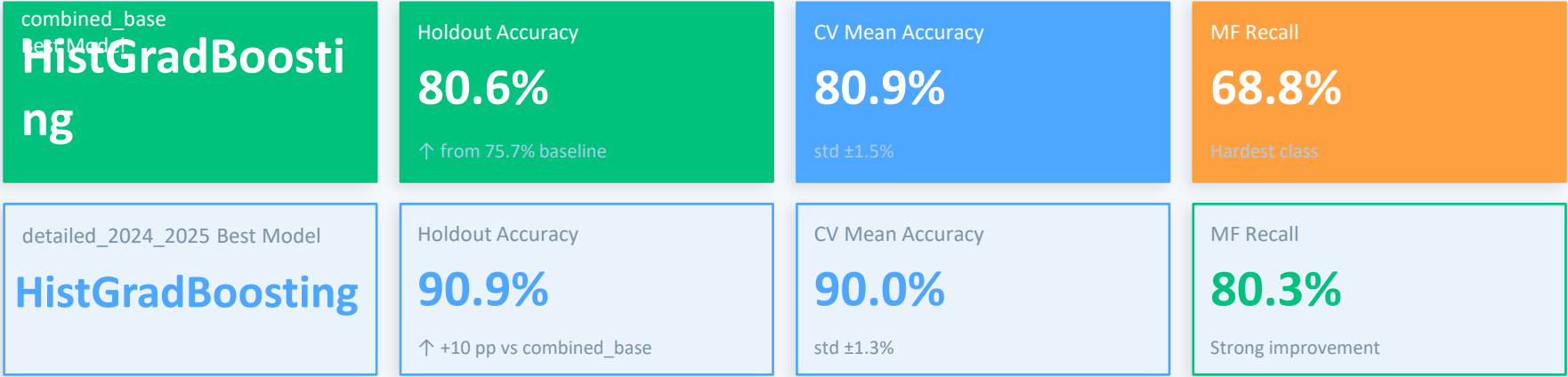
Aspect	Before (Original Notebook)	After (Refreshed Code)
Preprocessing	Inline string ops, manual dropna	Canonical pipeline via canonicalize_common_fields()
Encoding	Manual LabelEncoder per column	OneHotEncoder inside sklearn Pipeline
Imputation	dropna — rows deleted, data lost	Median imputation — no rows dropped
Scaling	StandardScaler re-instantiated per model	Model-specific scaling inside Pipeline
Model Selection	5 models, inconsistent CV	4 classifiers + 6 regressors with 5-fold StratifiedKfold
Regression Eval	Rounded preds → int, MSE only	Continuous; MAE, MSE, RMSE, R ² all reported + CV
Classification Eval	Accuracy only	Accuracy + Macro-F1 + per-class recall (MF focus)

Classification Results

Baseline (Original): RF Holdout Acc **0.7567** | RF CV Mean **0.7394** | *No Macro-F1 or recall reported*



Classification — Cross-Validation & Recall Breakdown



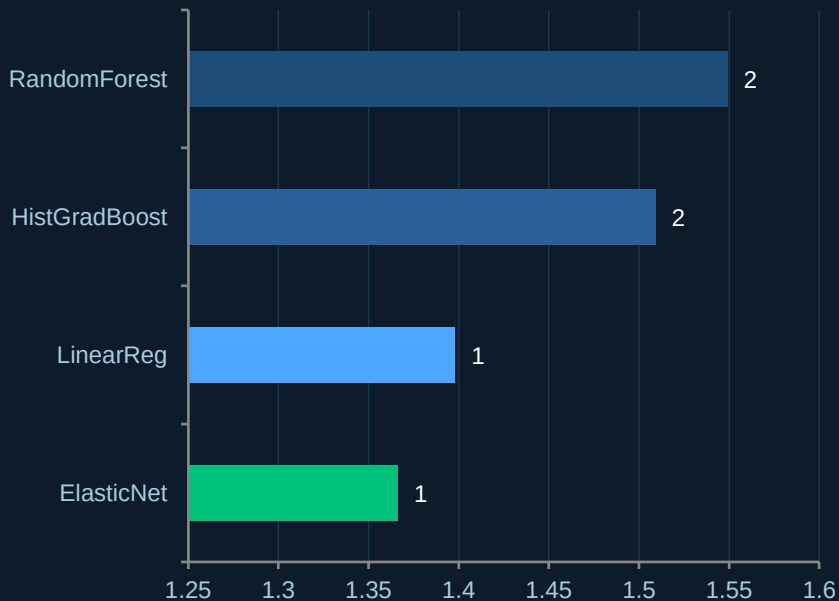
Model	Experiment	CV Acc Mean	CV F1 Mean	MF Recall CV
HistGradBoosting	combined_base	80.9%	84.1%	69.3%
RandomForest	combined_base	78.1%	81.6%	59.7%
LogisticReg	combined_base	78.8%	82.4%	66.1%
HistGradBoosting	detailed_2024_25	90.0%	91.1%	80.7%
RandomForest	detailed_2024_25	88.4%	89.9%	75.7%
LogisticReg	detailed_2024_25	88.7%	90.2%	80.3%

Regression Results

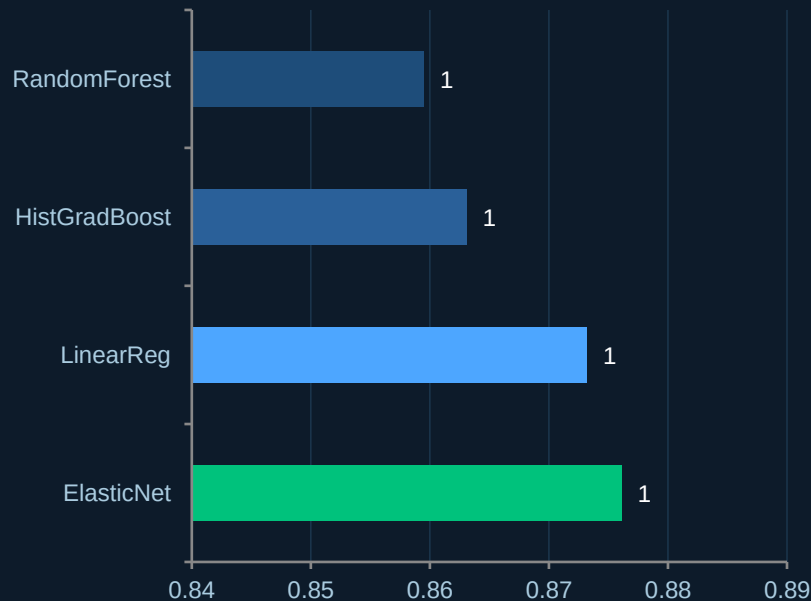
Predicting Goals Scored — Combined Base Dataset

Baseline (Original): LinearRegression (rounded preds) MSE **1.3892** R^2 **0.8625**

MSE (↓ Better)



R^2 Score (↑ Better)



Regression — Full Model Comparison

Best Model	MAE	MSE	R ²	CV R ² Mean
ElasticNet	0.7805	1.3664	0.8761	0.8694

Model	Holdout MAE	Holdout MSE	Holdout RMSE	Holdout R ²	CV R ² Mean	CV R ² Std
ElasticNet	0.7805	1.3664	1.1689	0.8761	0.8694	±0.0071
LinearRegression	0.7958	1.3982	1.1824	0.8732	0.8650	±0.0074
HistGradBoostRegressor	0.7672	1.5093	1.2285	0.8631	0.8555	±0.0101
RandomForestRegressor	0.7731	1.5493	1.2447	0.8595	0.8595	±0.0089
PoissonRegressor	1.1057	4.7178	2.1721	0.5721	0.5793	±0.0502
TweedieRegressor	1.3855	20.670	4.5465	-0.875	-0.644	±0.324

★ ElasticNet wins on MSE and R². TweedieRegressor not suited for this target distribution.

Before vs. After — At a Glance

Metric	Before	After	Δ Gain	Best Result
Classification Accuracy (RF holdout)	75.67%	80.63%	+4.96 pp	90.93% (detailed)
CV Mean Accuracy	73.94%	80.93%	+6.99 pp	90.03% (detailed)
Macro-F1	Not reported	0.8422	New metric	0.9175 (detailed)
MF Recall	Not reported	68.8%	New metric	80.3% (detailed)
Regression MSE	1.3892 (int)	1.3664 (cont.)	-1.6%	ElasticNet
Regression R ²	0.8625	0.8761	+0.0136	ElasticNet

What's Holding Us Back

Root causes of remaining classification error and regression limits

1. Noisy Hybrid Labels (~30%)

HIGH

29.7% of rows are mixed-position (FW,MF / DF,MF etc.). We collapse to first position → model trained on incomplete truths.

2. Midfield is the Hardest Class

HIGH

MF recall in combined_base: 68.8%. GK is near-perfect. Real challenge is separating MF from DF and FW at the boundaries.

3. Feature Gap in Combined Dataset

MED

combined_base only uses aligned shared columns. Rich 2024/25 features unavailable in older season → acc 80.6% vs 90.9%.

4. Position is Tactical, Not Statistical

MED

Formations, side-of-pitch, out-of-possession shape, match state — none appear in aggregate CSV stats.

5. Overlapping Roles Defy Clean Clusters

MED

Full-backs vs wide MFs. DMs vs CBs in build-up. AMs vs second strikers. Similar stats, different labels — theoretical ceiling.

6. Same-Season Benchmarking Confound

LOW

We predict position from same-season stats. Valid for benchmarking, but model partially learns role → stat entanglement.

Paths Forward

Recommended next steps to push accuracy and reliability further

Data

- ▶ Resolve hybrid labels — train a multi-label classifier or use probability outputs per position
- ▶ Collect more FBref seasons to enrich combined_base with more aligned shared features

Modelling

- ▶ Try multi-label approaches for hybrid positions rather than hard first-pos collapse
- ▶ Hyperparameter tuning via Optuna on HistGradBoosting for marginal accuracy gains

Features

- ▶ Per-position deviation features (how far above/below mean per role)
- ▶ Add defensive stats (tackles, pressures) to sharpen DF/MF classification boundary

Evaluation

- ▶ Confusion matrix heatmap per class — identify top misclassification pairs
- ▶ Calibration curves for regression — verify ElasticNet confidence intervals

Bottom Line



The original notebook was limited by feature design and evaluation quality — both are now fixed.



Cross-season combined data lifts classification accuracy safely from 75.7% → 80.6%.



The detailed 2024/25 experiment achieves 90.9% holdout accuracy with proper leakage removal.



Regression improved: ElasticNet delivers MSE 1.366 and R^2 0.876 on continuous (not rounded) predictions.



~30% hybrid labels, tactical ambiguity, and missing features form a natural accuracy ceiling.



Next focus: multi-label position modelling + deeper per-role feature engineering.